Bishal Deb

bishal@gonitsora.com
https://bishaldeb.com

Research Interests

Algebraic, Bijective and Enumerative Combinatorics

Research keywords: Coefficientwise Hankel total positivity, continued fractions, permutation statistics, dimer model

Employment and Internships

Mar 2025 - Present ◇ Postdoctoral researcher at
Yau Mathematical Sciences Center, Tsinghua University, Beijing, China
Supported by Tsinghua University Shuimu scholarship

Nov 2023 - Oct 2024 ◇ Postdoctoral researcher at
Laboratoire de Probabilités, Statistique et Modélisation, Sorbonne Université and Université Paris Cité, CNRS, Paris, France.
Supported by DIMERS project ANR-18-CE40-0033 funded by Agence Nationale de la Recherche (ANR, France).

Feb 2022 - Oct 2022 ◇ Data Engineer at Realm Inc., Part-time employee.

May 2018 - Jul 2018 ◇ Product Intern at Adobe Systems, Noida, India.
Project title: Topological Data Analysis
Supervisor: K. Balaji

May 2017 - Jul 2017 ◇ Product Intern at Adobe Systems, Noida.
Project title: Singular Learning Theory
Supervisor: K. Balaji

Education

Sep 2019 - Oct 2023 ◇ **PhD. in Mathematics, University College London (UCL), United Kingdom**
Funded by Dean's Scholarship and Departmental Teaching Assistantship
Dissertation title: Enumerative combinatorics, continued fractions and total positivity
Proquest link
Supervisor: Prof. Alan D. Sokal

Sep 2018 - Aug 2019 ◇ **Master 2 Informatique Fondamentale, Université Paris-Est Marne-la-Vallée (UPEM)*, France**
Résultat d'admission (Grade) 16.177/20
Funded by LabEx Bezout M2 scholarship (2018-19)
Thesis title: Combinatorics of Stammering Tableaux
Supervisors: Prof. Samuele Giraudo and Prof. Matthieu Josuat-Vergès

*now Université Gustave Eiffel

Aug 2017 - Jul 2018 ◇ **First year of MSc. in Computer Science, Chennai Mathematical Institute (CMI), India.** CGPA 9.43/10
Full scholarship from CMI

Aug 2014 - Jul 2017 ◇ **BSc. (Hons) in Mathematics and Computer Science, CMI.** CGPA 8.89/10
Funded by INSPIRE SHE Scholarship, Government of India

Publications

Journal Publications

[1] B. Deb, and A. D. Sokal; *Higher-order Stirling cycle and subset triangles: Total positivity, continued fractions and real-rootedness.* (54 pages). Accepted for publication in SIAM Journal on Discrete Mathematics.
arxiv preprint: https://www.arxiv.org/abs/2507.18959

[2] B. Deb; *Cyclic sieving phenomena via combinatorics of continued fraction.* (51 pages); In Discrete Mathematics, Volume 349, Issue 12 (2026), Article Number: 115284.
doi: 10.1016/j.disc.2026.115284

[3] V. Bitonti, B. Deb, and A. D. Sokal; *Thron-type continued fractions (T-fractions) for some classes of increasing trees*; In Electronic Journal of Combinatorics, Volume 33, Issue 1 (2026), Article number: P1.5. (68 pages).
URL: https://www.combinatorics.org/ojs/index.php/eljc/article/view/v33i1p5

[4] B. Deb; *Continued fractions using a Laguerre digraph interpretation of the Foata-Zeilberger bijection and its variants* (73 pages); In Combinatorial Theory, Volume 5, Issue 1 (March 2025).
doi: 10.5070/C65165020

[5] B. Deb, and A. D. Sokal; *Continued fractions for cycle-alternating permutations* (48 pages); In The Ramanujan Journal, (September 2024). doi: 10.1007/s11139-024-00905-7
URL: https://link.springer.com/article/10.1007/s11139-024-00905-7

[6] B. Deb, and A. D. Sokal; *Classical continued fractions for some multivariate polynomials generalizing the Genocchi and median Genocchi numbers*; In Advances in Applied Mathematics, Volume 161, (October 2024), Article number: 102756. (92 pages). doi: 10.1016/j.aam.2024.102756.
URL: https://www.sciencedirect.com/science/article/pii/S0196885824000885

[7] B. Deb, and A. D. Sokal; *A remark on continued fractions for permutations and D-permutations with a weight -1 per cycle*; In Electronic Journal of Combinatorics, Volume 31, Issue 2 (2024), Article number: P2.14. (34 pages). doi:10.37236/12149
URL: https://www.combinatorics.org/ojs/index.php/eljc/article/view/v31i2p14

[8] A. Clifton, B. Deb, Y. Huang, S. Spiro, and S. Yoo; *Continuously Increasing Subsequences of Random Multiset Permutations*; In European Journal of Combinatorics, Volume 110, 103708 (May 2023). (20 pages). doi:10.1016/j.ejc.2023.103708.
URL: https://www.sciencedirect.com/science/article/pii/S0195669823000252

[9] B. Deb; *Bijection between Increasing Binary Trees and Rook Placements on Double Staircases*; In Electronic Journal of Combinatorics, Volume 30, Issue 1 (2023), Article number: P1.10. (18 pages). doi:10.37236/10926.
URL: https://www.combinatorics.org/ojs/index.php/eljc/article/view/v30i1p10

Publications in Conference Proceedings

[10] A. Clifton, B. Deb, Y. Huang, S. Spiro, and S. Yoo; *Continuously Increasing Subsequences of Random Multiset Permutations*; In Séminaire Lotharingien de Combinatoire 86B (2022), Proceedings of the 34th International Conference on "Formal Power Series and Algebraic Combinatorics." URL: https://www.mat.univie.ac.at/ slc/wpapers/FPSAC2022/4.html

[11] X. Chen, B. Deb, A. Dyachenko, T. Gilmore, and A. D. Sokal; *Coefficientwise total positivity of some matrices defined by linear recurrences*; In Séminaire Lotharingien de Combinatoire 85B (2021), Proceedings of the 33rd International Conference on "Formal Power Series and Algebraic Combinatorics." URL: https://www.mat.univie.ac.at/ slc/wpapers/FPSAC2021/30.html

[12] B. Deb, A. Sarkar, N. Kumari, A. Rupela, P. Gupta, and Balaji K.; *Multimapper: Data Density Sensitive Topological Visualization*; IEEE International Conference on Data Mining Workshops (ICDMW), 2018. doi:10.1109/ICDMW.2018.00153

Patents

[13] A. Rupela, P. Gupta, N. Kumari, B. Deb, Balaji K., and A. Sarkar; *Generating varied-scale topological visualizations of multi-dimensional data*; U.S. Patent 11,100,127, issued August 24, 2021. https://patents.google.com/patent/US11100127B2/en

Preprints

[1] B. Deb; *Generalised hyperbolic polynomials: combinatorial interpretation and real-rootedness*. (18 pages). arxiv preprint: `https://arxiv.org/abs/2610.37931`.

[2] B. Deb, A. Dyachenko, M. Pétréolle, and A. D. Sokal; *Lattice paths and branched continued fractions III. Generalizations of the Laguerre, rook and Lah polynomials*. (86 pages arxiv, 89 pages in revision). arXiv preprint: `https://arxiv.org/abs/2312.11081`.

[3] B. Deb, and A. D. Sokal; *Coefficientwise Hankel-total positivity of the Schett polynomials*. (12 pages). arXiv preprint: https://arxiv.org/abs/2311.11747

[4] B. Deb; *Chromatic Polynomial and Heaps of Pieces*;
arXiv preprint: https://arxiv.org/abs/1902.02240

Talks and posters

Conference Presentations

- Coefficientwise Hankel total positivity: a survey, (June 22nd, 2026) at the Interactions between Combinatorics, Analysis and Statistical Physics: A conference in (belated) celebration of Alan Sokal's 70th birthday, University College London, London, UK.
- *A Laguerre digraph interpretation of a Biane-like bijection for D-permutations*, (June 7th, 2026) at the 2026 Seminar on Enumerative Combinatorics, The Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Qingdao, China.
- *A production matrix for the Schett polynomials via checkerboard exponential Riordan Arrays*, (June 21st, 2025) at the 10th International Conference on Riordan Arrays and Related Topics, Jiangsu Normal University, Xuzhou, China.
- *Coefficientwise Hankel-total Positivity of the Schett polynomials*, (June 2nd, 2025) poster presentation at Representation Theory, Integrable Systems and Related Topics 2025.

- *Continued fractions using a Laguerre digraph interpretation of the Foata–Zeilberger bijection*
  - (July 3rd, 2024) at the 30th British Combinatorial Conference
  - (September 5th, 2023) at International Conference on Enumerative Combinatorics and Applications 2023.
  - (July 27th, 2023) online poster presentation at AlCoVE: an Algebraic Combinatorics Virtual Expedition 2023.
  - (July 6th, 2023) at Permutation Patterns 2023.
- *A remark on continued fractions for permutations and D-permutations with a weight -1 per cycle*, (June 17th, 2024) online poster presentation at AlCoVE: an Algebraic Combinatorics Virtual Expedition 2024.
- *Laguerre digraphs and continued fractions*, (May 23, 2023) at Scottish Combinatorics Meeting 2023, University of Strathclyde, Glasgow, UK.
- *Continuously Increasing Subsequences of Random Multiset Permutations*, (July 18, 2022) poster presentation at 34th International Conference on Formal Power Series & Algebraic Combinatorics, Indian Institute of Science, Bangalore, India.
- *Multivariate continued fractions associated to Genocchi and median Genocchi numbers*, (June 9, 2022) at Heilbronn Meeting on Positivity Problems Associated to Permutation Patterns, Lancaster University, UK.
- *Combinatorics of continued fractions in the realm of permutations and Genocchi numbers*, (March 26, 2022) at Graduate Students Combinatorics Conference 2022
- *Total positivity of the Eulerian triangle: A big generalisation of Brenti's conjecture*,
  - (July 16th, 2021) British Early Career Mathematicians' Colloquium 2021,
  - (June 8th, 2021) at Young Researchers in Mathematics 2021,
  - (April 24th, 2021) at Graduate Students Combinatorics Conference 2021.

Seminar Presentations

2026:

- Coefficientwise Hankel total positivity: an introduction, (September 8th, 2026), at the Academy of Mathematics and Systems Science, Chinese Academy of Sciences.
- *Fock's dimer model on Speyer graphs: Let's draw train-tracks on crosses-and-wrenches graphs*, (May 23rd, 2026), at Postdoc Workshop, Yau Mathematical Sciences Center.
- *Hankel total positivity of the Schett polynomials: generalisations of continued fractions for Jacobian elliptic functions*, (February 9th, 2026), at the Indian Institute of Science, Bengaluru, India.
- *Continued fractions using a Laguerre digraph interpretation of the Foata–Zeilberger bijection and its variants*, (January 6th, 2026), at Center for Combinatorics, Nankai University, Tianjin, China.

2025:

- *Total positivity of three types: conjectures and observations*, (December 17th, 2025), at Dalian University of Technology, Dalian, China.
- *Coefficientwise total positivity of Hankel matrices in counting problems*, (November 15th, 2025), at Postdoc Workshop, Yau Mathematical Sciences Center.

- *Some bijections in the n! garden*, (April 9th, 2025) at Ahmedabad University.
- *Coefficientwise total positivity of Hankel matrices in counting problems*, (February 4th, 2025) at Al-Khwarizmi Applied Mathematics Webinar

2024:

- *Continued fractions using a Laguerre digraph interpretation of the Foata–Zeilberger bijection and its variants*
  - (September 23rd, 2024) at Laboratoire Bordelais de Recherche en Informatique, Bordeaux, France.
  - (April 16th, 2024) at the Institut Camille Jordan, Lyon, France.
  - (March 26th, 2024) at the Institut de Recherche en Informatique Fondamentale, Paris, France.
- *Coefficientwise total positivity of Hankel matrices in counting problems*,
  - (August 28th, 2024) at Ahmedabad University.
  - (June 7th, 2024) at Institut de Mathématiques de Jussieu-Paris Rive Gauche, Paris.
- *Coefficientwise Hankel-total positivity of the Laguerre polynomials*,
  - (June 14th, 2024) at Laboratoire de Probabilités, Statistique et Modélisation, Paris, France.
  - (May 28th, 2024) at Laboratoire d'informatique de Paris-Nord, Paris, France.
  - (May 24th, 2024) at University of Tours, Tours, France
- *Coefficientwise Hankel-total positivity of the Schett polynomials*, (February 27th, 2024) at Chennai Mathematical Institute, Chennai, India
- *Accumulation of zeroes for polynomials with combinatorial significance*, (February 23rd, 2024) at Combinatorics-related Open Problems Seminar (CROPS)

2023:

- *The "quadratic family" of continued fractions and combinatorial sequences*, (May 11th, 2023), Topics in Special Function and Number Theory (SFNT). (Invited)

2022:

- *Combinatorics and total positivity*, (August 23rd, 2022) Chennai Mathematical Institute, Chennai, India. (Invited)
- *Continued fractions and combinatorial sequences: factorials, Genocchi and median Genocchi numbers*, (August 17th, 2022) Institute of Mathematical Sciences, Chennai, India. (Invited)
- *Combinatorics of continued fractions in the realm of permutations and Genocchi numbers*, (April 20, 2022) at Graduate Online Combinatorics Colloquium. (Invited)
- *Combinatorial Hodge Theory*, (March 9, 2022) at UCL-LSE combinatorics reading group.

2021:

- *Analysing a strategy for a card guessing game via continuously increasing subsequences in multiset permutations*,
  - (November 18th, 2021) at Algebra, Representations, Combinatorics and Symmetric functions in India (ARCSIN).
  - (November 15th, 2021) at UCL Combinatorics Seminar series.

2020:

- *Total Positivity during Distressing Times*, (November 17, 2020) at Postgraduate Student Seminar Series, Department of Mathematics, UCL.
- *Combinatorics of Stammering Tableaux*, (July 31, 2022) at Online Weekly Research Seminar for Early Career Mathematicians from India. (Invited)

2019:

- *A Mathematical Pursuit of Happiness*, (September 14, 2019) outreach talk to mathematics undergraduates at Digboi College, Assam, India.
- *Chromatic Polynomials and Heaps of Pieces*, (May 20, 2019) at GT Combinatoire Énumérative et Algébrique, Laboratoire Bordelais de Recherche En Informatique, Université de Bordeaux, France. (Invited)

2017:

- *Chromatic Polynomials and Heaps of Pieces*, (March 14, 2017) at Institute of Mathematical Sciences, Chennai.

Conferences/Workshops Attended

In-person

- Interactions between Combinatorics, Analysis and Statistical Physics: A conference in (belated) celebration of Alan Sokal's 70th birthday, (June 2026), University College London, London, UK.
- 2026 Seminar on Enumerative Combinatorics, The Research Center for Mathematics and Interdisciplinary Sciences, June 2026, Shandong University, Qingdao, China.
- Representation Theory, Integrable Systems and Related Topics 2025, June 2026, Beijing Institute of Mathematical Sciences and Applications, Beijing, China.
- Representation Theory, Integrable Systems and Related Topics 2025, June 2025, Beijing Institute of Mathematical Sciences and Applications, Beijing, China.
- 10th International Conference on Riordan Arrays and Related Topics, June 2025, Jiangsu Normal University, Xuzhou, China.
- Formal Power Series and Algebraic Combinatorics 2024, July 2024, Ruhr-Universität Bochum, Bochum, Germany.
- British Combinatorial Conference 2024, July 2024, Queen Mary University of London, London, United Kingdom.
- At the crossroads of physics and mathematics : the joy of integrable combinatorics — A conference in the honor of Philippe Di Francesco's 60th birthday, June 2024, Institut de Physique Théorique, CEA Paris-Saclay, France.

- Journées ALEA 2024, March 2024, Centre International de Rencontres Mathématiques, Luminy, France.
- Workshop on 'Combinatorics, Algorithms, and Geometry', March 2024, Dresden, Germany.
- Workshop on Computer Algebra for Functional Equations in Combinatorics and Physics, December 2023, Institut Henri Poincare, Paris, France.
- Permutation Patterns 2023, University of Burgundy, Dijon, France.
- Scottish Combinatorics Meeting 2023, University of Strathclyde, Glasgow, UK.
- London Colloquia in Combinatorics 2023, London School of Economics, London, UK.
- Formal Power Series and Algebraic Combinatorics 2022, Indian Institute of Science, Bangalore, India.
- Algorithmic and Enumerative Combinatorics 2022, Technical University of Vienna, Austria.
- Early Career Researchers in Mathematics 2022, UCL, UK.
- Heilbronn Meeting on Positivity Problems Associated to Permutation Patterns, 2022, Lancaster University, UK.
- Göran Gustafsson Symposium 2022, Institut Mittag-Leffler, Sweden.
- Summer School on Geometric and Algebraic Combinatorics, 2019, Sorbonne Université at Jussieu, France.

Online

2022-23

- International Conference on Enumerative Combinatorics and Applications 2023
- AlCoVE: an Algebraic Combinatorics Virtual Expedition 2023

2021-22

- Graduate Student Combinatorics Conference 2022

2020-21

- Lattice Paths, Combinatorics and Interactions
- Permutation Patterns 2021
- Graduate Student Combinatorics Conference 2021
- Formal Power Series and Algebraic Combinatorics
- AlCoVE: an Algebraic Combinatorics Virtual Expedition
- Young Researchers in Mathematics 2021
- British Early Career Mathematicians Colloquium 2021

2019-20

- Algebraic Combinatorics Online Workshop (ACOW)
- AlCoVE: an Algebraic Combinatorics Virtual Expedition
- Formal Power Series and Algebraic Combinatorics 2020

Teaching Experience

- **Tutoring at YMSC** – Teaching assistant for the course Python Programming for Data and Analysis during fall 2025 and 2026.
- **Modules Tutored at UCL** – Analysis 1 and 2 (MATH0003 and MATH0004), Algebra 1 and 2 (MATH0005 and MATH0006), Further Linear Algebra (MATH0014), Mathematical Methods 3 (MATH0016).
- **Moderating Experience at UCL** – Online problem sessions for MATH0016.
- **Marking experience at UCL** – MATH0005, MATH0016, Combinatorial Optimisation (MATH0028), An Introduction to Mathematica (MATH0032), Computational Methods (MATH0058).
- Developer for UCL's STACK (System for Teaching and Assessment using a Computer algebra Kernel) question bank.
- **Marking experience at CMI** – Topics in Combinatorics (Spring 2018).

Skills

- **Programming Languages** - `Haskell, Java, Mathematica, Python, Sage, SQL`
- **Typesetting Languages** - LaTeX
- **Web designing** - *HTML, CSS*
- **Languages** : English (Fluent), Bengali (Native tongue), Hindi (Working proficiency), Assamese (Can read and speak), French (Beginners).

Other Professional Roles and Activities

- Reviewer for Annals of Combinatorics, Discrete Mathematics, Electronic Journal of Combinatorics, Journal of the London Mathematical Society, Formal Power Series and Algebraic Combinatorics (2024, 2025), Linear Algebra and its Applications, Journal of Combinatorics.
- Section Manager for Combinatorics, Frontiers of Science Awards 2026, 2027, International Congress of Basic Science.
- Organiser of Interactions between Combinatorics, Analysis and Statistical Physics: A conference in (belated) celebration of Alan Sokal's 70th birthday, (June 2026), University College London, London, UK.
- Organiser of ECRM 2022, an in-person conference for mathematics PhD students in UK organised by PhD students from Department of Mathematics, UCL.
- Research Student Representative at Department of Mathematics, UCL (September 2020 - May 2022).
- Editor of Gonit Sora, a bilingual web magazine.
- Organised a successful series of webinars (since June 2020) on behalf of Gonit Sora (Gonit Sora webinar page).
- Volunteered at London Mathematics Outreach.

- Joint Event Co-ordinator of Tessellate 2018, the academic and cultural festival of CMI.
- Organiser of Student Talks at CMI (2017-2018).
- Visharad (Graduate) in Tabla (Indian Percussion Instrument) from Bhatkhande Sangit Vidyapith, Lucknow, India. Senior Diploma in Art & Craft from Society of Art and Craft, West Bengal, India. Vocal music student at Darbar Academy (2021 - Present).

Awards, Grants and Honours

- Tsinghua University Shuimu Scholar (March 2025 - Present).
- Runner-up of the British Combinatorial Committee PhD Thesis Prize in Combinatorics. Citation available here: https://www.britishcombinatorics.org.uk/highlights.
- Mayer de Rothschild Studentship (2021-2022) at Department of Mathematics, UCL.
- Davenport Prize in Pure Mathematics (2020-2021) for best PhD student in Pure Mathematics for work done in the first year of research at Department of Mathematics, UCL.
- One of seven international students selected for LabEx Bezout M2 scholarship (2018-19).
- Recipient of INSPIRE SHE scholarship for being in the top 1% of Assam State board in class 12 exams.

Last updated on October 30, 2026.